

# Story of Geometry

Part 2





## About the Course

This course gives students a glimpse into how the ideas that guide our study of geometry were first discovered.



## Placement & Combining Tips

This course is intended to follow Part 1; however, the order may be reversed to facilitate combining students.



## Scheduling

| GRADE | SCHEDULE INFO.                                  | BOOKS                             |
|-------|-------------------------------------------------|-----------------------------------|
| 8     | Story of Geometry: Part 2<br>1 time/week 15 min | String, Straight-Edge, and Shadow |

### Sample Weekly View

| Day 1                     | Day 2 | Day 3 | Day 4                     | Day 5 |
|---------------------------|-------|-------|---------------------------|-------|
| Story of Geometry: Part 2 |       |       |                           |       |
|                           |       |       | Story of Geometry: Part 2 |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

### Story of Geometry: Part 2



String, Straight-Edge, and Shadow



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

[Click THIS text  
or scan the QR  
code for links.](#)



# Story of Geometry: Part 2

## How To Teach



### Recap

Connect back to previous lesson.



### Introduce

Optional: Use a picture, map, or engaging question to prepare the student to receive the idea from the day's lesson. Talk about one or two unknown words that will be crucial to understanding.



### Read

Read the new passage out loud or have the student read it orally or silently.



### Narrate

Have the student narrate orally or in writing.



### Discuss

- Give students an opportunity to share their thoughts on the lesson and any connections made.



### Practice

Optional: Attempt to recreate some of the problems from the stories.

# Story of Geometry: Part 2

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## Term 1

### WEEK 1 15m Story of Geometry: Part 2 - Lesson 1

*The Rules of the Game*

Materials: String, Straight-Edge, and Shadow

#### → READ, NARRATE, & DISCUSS

Ch.10 p.79-81 (UK 72-75) "With this" - "agreements and definitions."

- How was Thales' use of geometry similar to and different from the Egyptians'?

### WEEK 2 15m Story of Geometry: Part 2 - Lesson 2

*Circles: Terms, Definitions, & Rules*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.10 p.82-84 (UK 75-78) "We can hardly" - "than two angles."

#### → PRACTICE

Using your geometry tools, draw a line that is 2 inches long. Use this as the radius for a circle. What is the diameter? Using the bottom picture as a model, divide the circle into 6 equal parts.

Image Resource: Materials: String, Straight-Edge, and Shadow p.83 (UK 77)

### WEEK 3 15m Story of Geometry: Part 2 - Lesson 3

*Triangles: Terms, Definitions, & Rules*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.10 p.85-87 (UK 78-80) "Speaking of" - "180 degrees."

#### → PRACTICE

Using your geometry tools, construct a right triangle with a base that is 2 inches long. Construct a similar triangle with a base that is 4 inches long.

### WEEK 4 15m Story of Geometry: Part 2 - Lesson 4

*Lines: Terms, Definitions, & Rules*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.10 p.88-91 (UK 81-84) "Since two" - "Greek theoretical geometry!"

#### → PRACTICE

Look at the picture. What must be the measure of angle 1? [Use what you know about equilateral triangles.] What about angle 2? [Use what you know about right triangles.] How do you know? Can you figure out the measures of the unlabeled angles?

Image Resource: Materials: String, Straight-Edge, and Shadow p.90 (UK 83)

#### • IMPORTANT DATES

Pythagoras (c.570 BC-c.495 BC)

# Story of Geometry: Part 2

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## Term 1

- Draw or paint a picture that shows how Pythagoras determined that the Earth was round.

### WEEK 5 15m Story of Geometry: Part 2 - Lesson 5

*Pythagoras Studies with Thales*

Materials: String, Straight-Edge, and Shadow; World Wall Map

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.11 p.93-95 (UK 86-88) "The early story" - "also was spherical."

#### → LOCATE

Map Resource: World Wall Map

Map Link: Croton

Look at the map and find Croton. Then find it on your wall map.

### WEEK 6 15m Story of Geometry: Part 2 - Lesson 6

*Pythagoras and His Followers*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.11 p.96-97 (UK 88-90) "And the radiant" - "meant mathematics."

### WEEK 7 15m Story of Geometry: Part 2 - Lesson 7

*A Famous Theorem*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.12 p.98-99 (UK 91-92) "The most famous" - "Secret Brotherhood."

- Copy the tile illustration on p.101 (UK 93). Pretend you are Pythagoras explaining this truth to your new initiates.

### WEEK 8 15m Story of Geometry: Part 2 - Lesson 8

*Great Mathematical Discoveries*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.12 p.100-101 (UK 92-94) "The very term" - "hypotenuse."

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## Term 1

### WEEK 9 15m Story of Geometry: Part 2 - Lesson 9

*A Rigorous Proof*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.12 p.102-103 (UK 94-95) "Again he waited" - "showed plainly."

### WEEK 10 15m Story of Geometry: Part 2 - Lesson 10

*Pythagoras' Theorem Defined*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.12 p.104-105 (UK 96-97) "Look again!" - "sacrifice renowned!"

#### → VIEW

Video on Pythagorean Theorem

∞ Video Link: Pythagorean Theorem

#### → PRACTICE

Construct your own 3-4-5 right triangle. Use cubes, tiles, or graph paper to prove the Pythagorean Theorem.

### WEEK 11 15m Story of Geometry: Part 2 - Lesson 11

*Catch-Up Day*

Materials: String, Straight-Edge, and Shadow

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 12 15m Story of Geometry: Part 2 - Lesson 12

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

Questions will come from:

String, Straight-Edge, and Shadow

# Story of Geometry: Part 2

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## Term 2

### WEEK 1 15m Story of Geometry: Part 2 - Lesson 13

*Dice of the Gods*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.13 p.106-107 (UK 98-99) "As time went on" - "geometric figure."

### WEEK 2 15m Story of Geometry: Part 2 - Lesson 14

*Regular Solids*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.13 p.108-109 (UK 99-101) "Probably the next" - "of the universe."

### WEEK 3 15m Story of Geometry: Part 2 - Lesson 15

*Building Shapes*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.13 p.110-111 (UK 101-103) "The cube" - "triangular faces."

### WEEK 4 15m Story of Geometry: Part 2 - Lesson 16

*The Dodecahedron*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.13 p.112-114 (UK 104-106) "The Dodecahedron" - "of the gods."

### WEEK 5 15m Story of Geometry: Part 2 - Lesson 17

*The Unspeakable Tragedy*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.14 p.115-117 (UK 107-109) "Before the Secret" - "on surest ground."

### WEEK 6 15m Story of Geometry: Part 2 - Lesson 18

*Whole Numbers & Geometric Shapes*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### • IMPORTANT DATES

Hippasus of Metapontum (c.500 BC)



# Story of Geometry: Part 2

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## Term 2

### → READ, NARRATE, & DISCUSS

Ch.14 p.118-119 (UK 109-110) "Numbers, they" - "disastrous experiment."

### → PRACTICE

Look at the pictures. What would be the next square number? What would be the next triangular number? Draw pictures.

📄 Image Resource: Materials: String, Straight-Edge, and Shadow p.118-119 (UK 110)

### WEEK 7 📅 15m Story of Geometry: Part 2 - Lesson 19

*Triangles & Irrational Number*

📄 Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.14 p.119-122 (UK 111-113) "Presiding was" - "discovery of irrationals."

### • IMPORTANT DATES

Plato (c.428 BC-c.348 BC)

Plato's Academy (est. c.387 BC)

### WEEK 8 📅 15m Story of Geometry: Part 2 - Lesson 20

*The Golden Age & the Golden Mean*

📄 Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.15 p.124-125 (UK 116-117) "The second half of" - "partly a Pythagorean."

### • IMPORTANT DATES

Eudoxus of Cnidus (c.390 BC-c.337 BC)

### WEEK 9 📅 15m Story of Geometry: Part 2 - Lesson 21

*Plato's Academy*

📄 Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.15 p.125-127 (UK 117-119) "When his teacher" - "then in vogue."

- Explain how Greek architects used a semicircle to construct a rectangle.

### WEEK 10 📅 15m Story of Geometry: Part 2 - Lesson 22

*Architecture: String & Straight-Edge*

📄 Materials: String, Straight-Edge, and Shadow

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

Ch.15 p.127-128 (UK 119-120) "Like many Greek" - "former master Plato."

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## Term 2

WEEK 11

15m Story of Geometry: Part 2 - Lesson 23

Proportion & Geometric Quantities

Materials: String, Straight-Edge, and Shadow

→ RECAP

What did you read last time?

→ READ, NARRATE, & DISCUSS

Ch.15 p.129-130 (UK 120-121) "This time, Eudoxus" - "this simplest Form."

WEEK 12

15m Story of Geometry: Part 2 - Lesson 24

Exam Week

→ EXAMS

Answer question(s) related to course.

Questions will come from:

String, Straight-Edge, and Shadow

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## Term 3

### WEEK 1 15m Story of Geometry: Part 2 - Lesson 25

*Right Triangles*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.15 p.131-133 (UK 121-123) "Consider only" - "solved the problem!"

### WEEK 2 15m Story of Geometry: Part 2 - Lesson 26

*The Root Five*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.15 p.133-134 (UK 123-125) "But sir, protested" - "use of it."

### WEEK 3 15m Story of Geometry: Part 2 - Lesson 27

*Proportions of the Human Body*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.15 p.135-136 (UK 125-126) "The facade of" - "Divine Section."

### WEEK 4 15m Story of Geometry: Part 2 - Lesson 28

*Geometric Construction of the Pentagon*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.15 p.137-138 (UK 126-128) "Yet the immediate" - "to come."

- Explain how the lifting of the "antimechanical taboo" led to advancement in many areas.

#### • IMPORTANT DATES

Aristotle (385 BC-323 BC)

The Lyceum (est. 335 BC)

Alexander the Great (356 BC-323 BC)

### WEEK 5 15m Story of Geometry: Part 2 - Lesson 29

*A Royal Road After All*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.16 p.139-141 (UK 129-131) "During the 4th" - "tackles and gears."

#### • IMPORTANT DATES

Archimedes of Syracuse (287 BC-212 BC)

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## Term 3

### WEEK 6 15m Story of Geometry: Part 2 - Lesson 30

*Archimedes of Syracuse*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.16 p.141-143 (UK 131-133) "The most illustrious" - "definitely proved."

### WEEK 7 15m Story of Geometry: Part 2 - Lesson 31

*Three Famous Puzzles*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.16 p.144-148 (UK 133-138) "So from the" - "pinnacle of its own."

#### • IMPORTANT DATES

Eratosthenes Measures Circumference of Earth (c.250 BC)

### WEEK 8 15m Story of Geometry: Part 2 - Lesson 32

*The Whole, Round Earth*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.17 p.149-151 (UK 139-142) "Our story of ancient" - "of the earth."

### WEEK 9 15m Story of Geometry: Part 2 - Lesson 33

*The Circumference of the Earth*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.17 p.152-153 (UK 143) "But the sight" - "desperately needed."

#### • IMPORTANT DATES

Euclid Publishes "Elements" (c.300 BC)

### WEEK 10 15m Story of Geometry: Part 2 - Lesson 34

*Euclid's Elements*

Materials: String, Straight-Edge, and Shadow

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

Ch.17 p.154-155 (UK 143-145) "Earlier Greeks had" - "and the shadow."

### WEEK 11 15m Story of Geometry: Part 2 - Lesson 35

*Catch-Up Day*

Materials: String, Straight-Edge, and Shadow

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore

# Story of Geometry: Part 2

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| Term 3                                                                                                                                                                                                                                                           |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Extra Helpings.                                                                                                                                                                                                                                                  |  |
| <div>WEEK 12</div> <div><div></div> 15m Story of Geometry: Part 2 - Lesson 36</div> <div>Exam Week</div> <div><div>→ EXAMS</div><div>Answer question(s) related to course.</div><div>Questions will come from:<br/>String, Straight-Edge, and Shadow</div></div> |  |

# Story of Geometry: Part 2

## Examination

- Term 1** · String, Straight-Edge, and Shadow: Tell what you know of the Pythagorean Theorem.
- Term 2** · String, Straight-Edge, and Shadow: Tell what you know of the Golden Mean.
- Term 3** · String, Straight-Edge, and Shadow: Tell about some of the geometrical feats that the Greeks discovered.